

Muhammad Saad

imsaad.work@gmail.com | 03214589835 | github.com/Im-Saad08 | Taxila, Pakistan

Professional Summary

Computer Engineering student with hands-on experience in data visualizations, embedded systems, and computer vision-based projects. Proven ability to develop practical and innovative solutions through academic and personal projects, including Python applications, microcontroller-based systems, and Python-based GUI applications. Skilled in Python, C++, OpenCV, and embedded programming, with a strong passion for software development, data analytics, and innovative engineering solutions.

Education

BSc Computer Engineering

2022 – Present | I-12, Islamabad

FSc Pre-Engineering

2020 – 2022 | PMO Colony, Taxila Cantt

Matriculation

2018 – 2020 | PMO Colony, Taxila Cantt

National University of Technology, Islamabad

Iqbal Campus, Jinnah Education System

Iqbal Campus, Jinnah Education System

Skills

Python, C++, MATLAB, Microsoft Excel, SQL, MySQL, Microchip PIC16F877A, Intel 8051, VS Code, MPLAB X IDE, UART, Tkinter, Pygame, OpenCV, NumPy, Data Visualization, Image Processing Fundamentals

Projects

Digital Thermometer and Humidity Monitor

Developed a firmware for sensor communication, data validation and hardware timing control.

Technologies: PIC16F877A, DHT22 Sensor

Hand Gesture Recognition System

Applied image preprocessing trained the model on a custom dataset, achieving 94.3% accuracy for 20+ gestures.

Technologies: Python, OpenCV, TensorFlow, MediaPipe

Smart Solar Powered Electric Bicycle

Proposed a solar charging station with calculations as a conceptual addition, enhancing project management and technical skills.

Technologies: Lithium-Ion Battery, BLDC Hub Motor, BLDC Motor Controller, Solar Charge Controllers

Laser-based Bidirectional Data Transmission

Integrated with Arduino UNO for precise control and signal processing, minimizing signal loss and ensuring efficient, bidirectional communication

Technologies: Arduino UNO, Laser Module, Solar Panel, PC

4 Way Traffic Light Controller

Built the hardware logic focusing on wiring and circuit design to simulate dynamic signal control for traffic management.

Technologies: LEDs, IC 7486, IC 7408, Veroboard

Interests

Data Analysis • Python Automation • Image Processing

Experience

Coordinator

Jehaad for Zero Thalassemia, JZT Pakistan
Sep 2024 – Present | NUTECH, Islamabad

Leading the university chapter focused on thalassemia prevention through awareness campaigns and student-led community outreach initiatives. Responsible for planning, organizing, and executing awareness activities, coordinating team members, and managing event operations within the university. Actively contributing to health education efforts while strengthening leadership, communication, and organizational skills through impactful social initiatives.

President

Global Youth Forum for Health and Awareness, GYFHA NUTECH
Feb 2026 – Present | NUTECH, Islamabad

Leading a student organization focused on health awareness initiatives, currently engaged in thalassemia prevention campaigns along with broader disease awareness programs. Responsible for organizing seminars, coordinating teams, and managing event planning and execution. Developing strong leadership, management, and problem-solving skills through active involvement in university-level health awareness and outreach activities.

Certifications

All Pakistan Minorities Heritage Photo Contest

Higher Education Commission
Apr 2023

Nexus of Mentors & Emerging Leaders

Jehaad for Zero Thalassemia, Pakistan
Jul 2025

Leadership Aptitude Management Program

Jehaad for Zero Thalassemia, Pakistan
May 2025

PIVOT 2.0 - Train the Trainer Program

Jehaad for Zero Thalassemia, Pakistan
Dec 2025

NUTECH IEEE Fest

IEEE Student Branch NUTECH
Dec 2025

Awards

Thalassemia Awareness

Higher Education Commission, Global Youth Forum for Health and Awareness
May 2026

Recognized for valuable contribution to Thalassemia Awareness

Languages

Urdu (Native) • English (Professional)